

INDU 211: Introduction to Production & Manufacturing Systems

Problem Solution Guide 06: Linear Programming Production Modeling

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[!NOTE]

- **YouTube Video Tutorial:** [Watch Video 6: INDU 211 - Modeling Problem](#)
- **Video ID:** `Rwc_f6IzUQk` · **Duration:** 18:11
- **Target Exam Scope:** ❏ **FINAL EXAM (Weeks 7–12, Chapters 14, 15, 8, 17)**
- **Curriculum Context:** Chapter 14 — Deterministic Operations Research Models

1. Problem Description & Operational Data

A pharmaceutical manufacturing plant produces three commercial prescription drugs: **Drug A, Drug B, and Drug C**. The production process involves raw active ingredients, limited machine tool hours, market contract minimums, and strict proportional batch blending requirements:

- **Raw Active Ingredients:**
 - **Ingredient 1:** Maximum available supply = **4,000** grams.
 - **Ingredient 2:** Maximum available supply = **6,000** grams.

Active Ingredient	Drug A (x_A)	Drug B (x_B)	Drug C (x_C)	Supply Limit
Ingredient 1	20 g/unit	33 g/unit	50 g/unit	$\leq 4,000$ g
Ingredient 2	50 g/unit	20 g/unit	60 g/unit	$\leq 6,000$ g

- **Machine Time Availability:**
 - One unit of Drug A requires **twice** the machine time of Drug B ($t_A = 2t_B$).
 - One unit of Drug A requires **three times** the machine time of Drug C ($t_A = 3t_C$).
 - Total machine time available in the factory is sufficient to produce **1,500 units of Drug A** if 100% dedicated to Drug A.
- **Minimum Market Demand Thresholds:**
 - Demand for Drug A $\geq$ **300** units.
 - Demand for Drug B $\geq$ **400** units.
 - Demand for Drug C $\geq$ **250** units.
- **Proportional Production Blend Ratio:**

- Production volumes must be maintained in the exact ratio:

$$x_A : x_B : x_C = 2 : 1 : 3$$

- **Profit Contribution Margins:**
 - Unit Profit for Drug A = **\$40 / unit**
 - Unit Profit for Drug B = **\$30 / unit**
 - Unit Profit for Drug C = **\$60 / unit**

2. Complete Mathematical LP Model Formulation

Step 1: Decision Variables

Let:

- x_A = Number of units of Drug A produced
- x_B = Number of units of Drug B produced
- x_C = Number of units of Drug C produced

Step 2: Objective Function

Maximize total company profit:

$$\text{Maximize } Z = 40x_A + 30x_B + 60x_C$$

Step 3: Raw Active Ingredient Constraints

- **Ingredient 1 Usage:**

$$20x_A + 33x_B + 50x_C \leq 4000$$

- **Ingredient 2 Usage:**

$$50x_A + 20x_B + 60x_C \leq 6000$$

Step 4: Machine Hour Capacity Constraint

- **Relative Time Equivalence Derivation:** Because $t_A = 2t_B$ and $t_A = 3t_C$, producing 1 unit of Drug B consumes only 0.5 equivalent units of Drug A machine capacity, and 1 unit of Drug C consumes $\frac{1}{3}$ equivalent units of Drug A capacity.
- **Capacity Inequality:**

$$x_A + 0.5x_B + \frac{1}{3}x_C \leq 1500$$

(Or in integer form: $6x_A + 3x_B + 2x_C \leq 9000$)

Step 5: Minimum Market Demand Constraints

To fulfill contracted customer shipments:

$$x_A \geq 300$$

$$x_B \geq 400$$

$$x_C \geq 250$$

Step 6: Proportional Production Ratio Constraints

The production quantities must satisfy the ratio $x_A : x_B : x_C = 2 : 1 : 3$:

$$\frac{x_A}{2} = \frac{x_B}{1} = \frac{x_C}{3}$$

This breaks into two linear equality equations:

$$1. \frac{x_A}{2} = x_B \implies x_A - 2x_B = 0$$

$$2. \frac{x_C}{3} = x_B \implies 3x_B - x_C = 0$$

Step 7: Non-Negativity Constraints

$$x_A \geq 0, \quad x_B \geq 0, \quad x_C \geq 0$$